



Personal Portfolio Web Application

SWE363 – Web Engineering / Assignment 4

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<https://haidar-aldahan.github.io/202256620-HaidarAlDahan-assignment4/>

What Did I Build?

*A fully deployed personal portfolio website — built from scratch with **HTML, CSS, and Vanilla JavaScript**. No frameworks. No shortcuts. Live on GitHub Pages.*

About

Personal introduction and background

Skills

Technical competencies displayed visually

Projects

Filterable, searchable project showcase

GitHub

Live repos via REST API

Contact

Working email form via EmailJS

Key Features



Responsive Design

Fluid layouts for mobile, tablet, and desktop



Dark / Light Theme

Instant toggle with persistent preference



Project Filtering & Search

Filter, search, and sort projects dynamically



Live GitHub API

Real-time repository data via REST API



EmailJS Contact Form

Real email delivery — no backend required



Scroll Animations

Typing hero effect, scroll reveals, back-to-top button

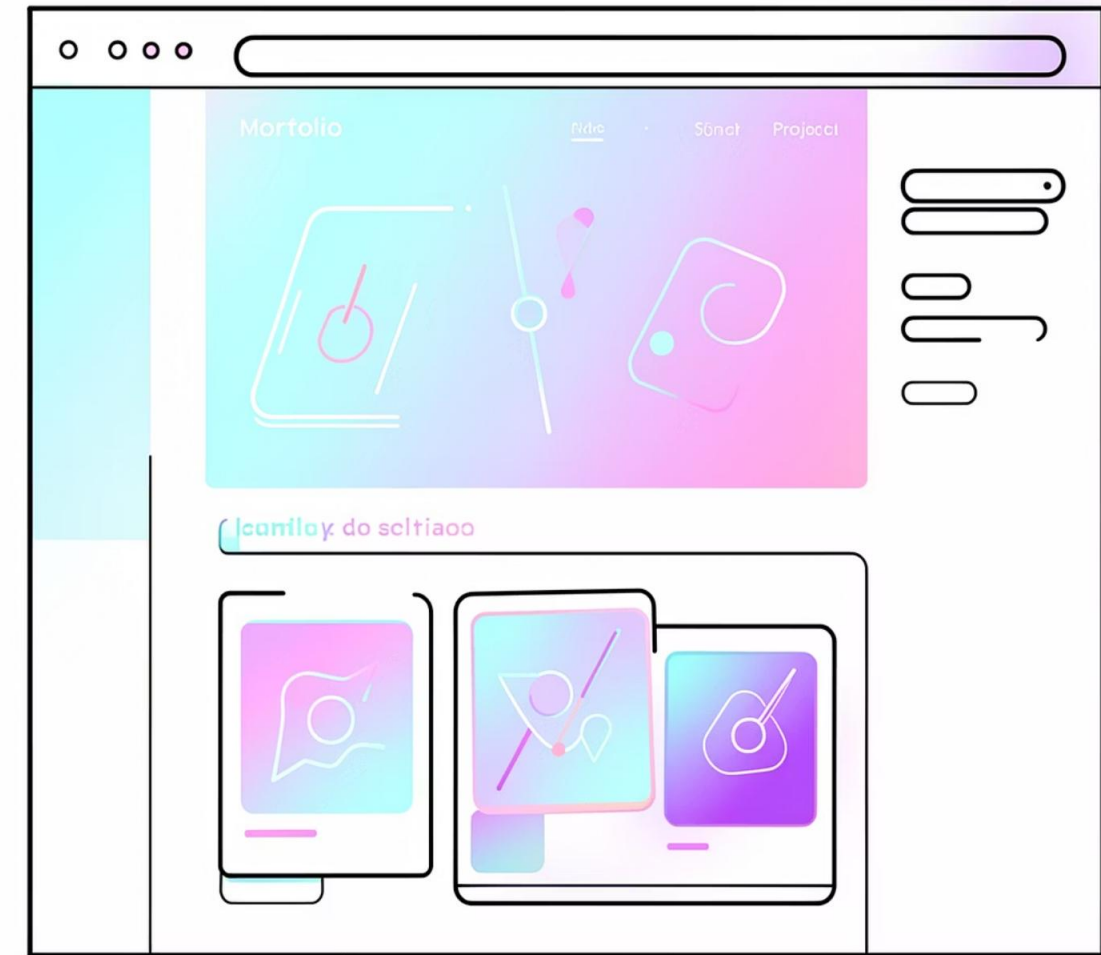
Live Demonstration

Switching to live site →

Explore the fully deployed portfolio live on GitHub Pages. Navigate all sections, toggle themes, filter projects, and test the contact form.

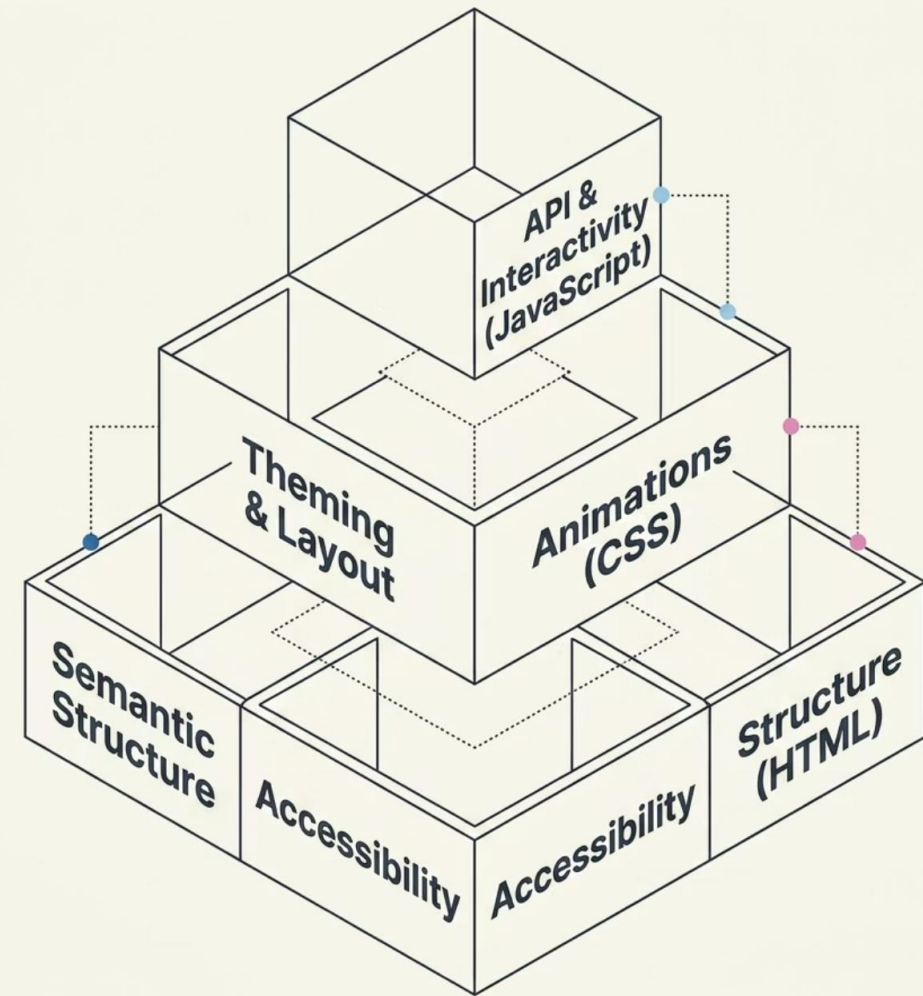


<https://haider-aldahan.github.io/202256620-HaiderAlDahan-assignment4/>



How It's Built

<i>index.html</i>	<i>Semantic structure & accessibility</i>
<i>css/styles.css</i>	<i>Theming, layout, animations</i>
<i>js/script.js</i>	<i>Interactivity, API calls, EmailJS</i>



📄 No frameworks — pure HTML, CSS, JS + GitHub API + EmailJS

AI Tools Used

Claude (Anthropic)

- *EmailJS service config, template variables & JS integration*
- *Debugged silent email failures (mismatched variable names)*
- *IntersectionObserver scroll animation logic*
- *Typewriter effect with role cycling & timing*
- *Resolved scroll animation conflicts with project cards*
- *Back-to-top button scroll event & CSS transition*
- *Code structure review & organization suggestions*
- *README, AI report & technical documentation drafts*

ChatGPT

- *Cross-checked debouncing implementation*
- *Validated GitHub API fetch & error handling*
- *Suggested back-to-top scroll threshold*
- *Alternative CSS animation approaches for comparison*

 *All AI output was reviewed, tested, and manually modified — AI assisted, never replaced my understanding.*

Technical Deep Dive



✓ Resolved

- *EmailJS variable mismatch — fixed manually*
- *Scroll animation conflicts — resolved with separate CSS classes*
- *GitHub API errors — handled with graceful fallback messages*



⚠ Unresolved

- *No auto-reply to contact form visitors*
- *GitHub API rate limit (60 req/hr without auth)*



🔭 Future Plans

- *Auto-reply via EmailJS configuration*
- *GitHub OAuth token for higher rate limits*
- *Blog / writing section*

What I Achieved

- Fully deployed on GitHub Pages
Live, accessible, and production-ready
- Responsive design without frameworks
Mobile-first CSS built from scratch
- Two real external APIs integrated
GitHub REST API + EmailJS
- Working email system — no backend
Serverless contact form via EmailJS
- Responsible AI-assisted development
AI as a learning tool, not a replacement

This portfolio is mine to keep, improve, and share proudly.